Nikhilesh Vijay Chirutala

Generative AI Developer

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SUMMARY

Generative AI Developer with experience building AI-enabled application workflows using Python, FastAPI, LLM integration, Retrieval-Augmented Generation (RAG), prompt design, and tool-calling patterns. Experienced in connecting AI capabilities with application data, APIs, and business workflows across fintech and CRM use cases. Strong foundation in backend development, REST APIs, data integration, authentication, and production support.

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Backend & Frameworks: FastAPI, Flask, REST APIs, API Development, Backend Automation

Frontend: React.js, Next.js, Tailwind CSS, Bootstrap, Jinja2, Microservices

AI & LLM: Generative AI Applications, Retrieval-Augmented Generation (RAG), LLM Integration, MCP Integration, Prompt Engineering

Databases: MongoDB, PostgreSQL, Neon DB, SQL

Cloud & DevOps: AWS, Amazon S3, Vercel, GitHub Actions, CI/CD, Linux

APIs & Integrations: Twilio, GitHub API, REST APIs, Postman, API Integration

Security & Authentication: Authentication, Authorization, RBAC, ACL, JWT

Testing & Tools: Postman, Jest, Unit Testing, Git, GitHub, Debugging

WORK EXPERIENCE

Software Engineer · **CA One Tech Cloud Inc.** Apr 2026 – Present | Bengaluru, India

Client: Convoso — Cloud Contact Center Platform *Python, FastAPI, AWS, Amazon S3, MySQL, LLM APIs, RAG, Datadog*

- Designed a GenAI-assisted call intelligence workflow that summarizes call transcripts and surfaces key topics for contact-center users, using an LLM-backed service integrated with existing Python APIs.
- Prototyped retrieval over approved call and account context, retrieving relevant records before generating grounded responses to agent questions and reducing reliance on manual searches.
- Implemented prompt templates and structured-output validation for call summaries, intent highlights, and follow-up suggestions, with application-level checks for incomplete or malformed model responses.
- Integrated model inference into asynchronous Python service workflows, including request validation, timeout handling, retry policies, and persistence of generated results for downstream application use.
- Added observability for AI-enabled service paths, tracking request failures, latency, and processing outcomes through dashboards and error reporting to support debugging and operational review.
- Designed human-review and fallback paths for uncertain or unavailable model responses, keeping generated content separate from source records and avoiding unsupported automatic actions.
- Evaluated representative prompts and sample inputs to check summary relevance, consistency with supplied context, and handling of empty or noisy transcripts.
- Worked across API, storage, and production-support layers to troubleshoot failures in the illustrative AI processing pipeline and maintain reliable integration with contact-center workflows.

Software Engineer · **Mantra Technologies** Oct 2023 – Mar 2026 | Hyderabad, India

- Built AI-assisted application workflows connecting language-model responses with structured product data and existing business operations through Python services and REST APIs.
- Developed a retrieval-augmented question-answering prototype over application records, combining relevant retrieved context with user questions to generate context-aware responses.
- Created tool-calling interfaces for selected application operations, allowing an AI assistant to request approved reads or actions through explicit schemas and backend authorization checks.
- Designed prompt templates and response parsing for structured outputs, with validation and fallback behavior for missing fields, malformed results, and unsupported requests.
- Integrated AI features with frontend flows and backend APIs, including loading and error states, input validation, and persistence of relevant application results.
- Tested AI workflows against representative user questions and edge cases, reviewing relevance, groundedness, tool selection, and whether responses followed the requested output format.
- Implemented authentication and access-control checks around AI-connected application operations so model-generated requests could not bypass existing permissions.

- Investigated integration issues across model APIs, backend services, and application data, documenting limitations and improving reliability through defensive error handling.

Key Projects:

AI Bull — GenAI-Enabled Trading & Investment Platform *React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM*

- Developed an AI-powered trading and investment platform with portfolio aggregation, enabling users to consolidate holdings from multiple Demat accounts such as Upstox, Zerodha, and Groww into a unified view.
- Built portfolio workflows for tracking positions, stocks, and mutual funds, along with detailed reports and investment-level information across aggregated accounts.
- Developed stock and mutual fund screeners with configurable filtering capabilities and detailed asset pages for analyzing individual investments and their underlying information.
- Designed and implemented a backtesting engine for evaluating trading strategies against historical market data and analyzing strategy performance.
- Implemented a Retrieval-Augmented Generation (RAG)-based chatbot that combines portfolio and financial information as context to answer portfolio-related questions and provide contextual investment insights.
- Built Python and FastAPI APIs and business logic connecting portfolio aggregation, investment analysis, screening, and backtesting workflows with the platform's AI-enabled user experiences.

CRM.io — CRM Platform with AI-Assisted Workflows *FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, WebSockets, AWS S3, AWS Lambda, FCM, MCP*

- Developed backend modules and REST APIs for an AI-driven CRM using Python and FastAPI, implementing business logic and workflows across Contacts, Tasks, Deals, Documents, communication, and account management.
- Built the Contacts management experience with inline editing, global search across CRM modules, access to related Tasks, Deals, and Documents, and Google Sheets integration for importing contact records in bulk.
- Developed a real-time CRM dialer integrated with Twilio and an internal SIP-based telephony application through WebSockets, with AI-generated call scripts available to support relevant customer conversations.
- Implemented a CRM-wide soft-delete and Trash workflow with a 30-day retention period, replacing hard-delete behavior and using an AWS Lambda service to permanently remove records after the retention period.
- Built document management workflows using Amazon S3, including browser-to-S3 client uploads and storage of documents associated with CRM records.
- Developed a task cadence system that allows users to create sequences of tasks as a single workflow and progress through scheduled follow-ups, with Firebase Cloud Messaging (FCM) used for task notifications.
- Built account-level reporting dashboards and an MCP integration exposing supported CRM capabilities as tools, enabling an AI bot to retrieve CRM information and invoke available operations through a defined tool interface.

PERSONAL PROJECTS

JS Studio — Online JavaScript Playground *Next.js, GitHub API, Vercel*

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel.

AnyScrum — Project Management Platform *Next.js, Neon DB, Clerk, Vercel*

- Developed a project management platform for projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

EDUCATION

B.Tech — Electrical & Electronics Engineering 2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur